压缩感知算法

残差 $r = y - Ax$ L 最大 k 下标

负梯度 $g = A^H r$ H 保留最大 k 元素

$$\frac{\partial}{\partial x}\|y - Ax\|_2^2 = -A^H(y - Ax)$$

贪心算法

MP $n = L(g) = \operatorname{argmax} \vec{g}_N$

$$x_n = x_n + \frac{1}{\|a_n\|} g_n$$

OMP $S = S \cup L(g)$

$$x = \operatorname{argmin}\{\|y - Ax\|,\ \operatorname{supp}(x) \subset S\} = A_S^{\dagger} y$$

Threshold 算法

Basic Threshold $S = L_k(g)$

$$x = \operatorname{argmin}\{\|y - Ax\|,\ \operatorname{supp}(x) \subset S\} = A_S^{\dagger} y$$

IHP $x = H_k(x + g)$

HTP $S = L_k(x + g)$

$$x = \operatorname{argmin}\{\|y - Ax\|,\ \operatorname{supp}(x) \subset S\} = A_S^{\dagger} y$$

OT (Optimal k-Threshold), ROT (Relaxation)

$$\tilde{x} = x + g$$

$$w = \operatorname{argmin}\{\|y - A(\tilde{x} \odot w)\|,\ 1^T w = k,\ w_m = 0/1\} \quad \text{OT}$$

$$x = \tilde{x} \odot w \qquad w \in [0, 1] \quad \text{ROT}$$

PLA (Projected Landweber Algorithm) $x = P_u(x + \alpha g)$

CoSaMP $S = S \cup L_{2k}(g)$

$$x = \operatorname{argmin}\{\|y - Ax\|,\ \operatorname{supp}(x) \subset S\} = A_S^{\dagger} y$$

$$x = H_k(x)$$